INTERNATIONAL GCSE MATHEMATICS A · LINEAR 4MA1 HIGHER

Expected Paper ELT/4MA1X Mark Scheme

Step by step, with the marks



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M = method mark, A = accuracy mark (needs the method), B = mark for a correct value on its own. Questions 19–26 also show THE IDEA: the one thought that unlocks them.

ANSWERS AT A GLANCE

1	shown: $\frac{8}{3} \times \frac{9}{8} = 3$
2	1.44% decrease
3	$2^3 \times 3^2 \times 7$; 2520
4	23, 12, 16, 9; $\frac{13}{20}$
5	$x = 4$, $y = 2$
6	10.9 km; 245°
7	70 km/h
8	$12x^2 + 7x - 10$; $6xy(3x - 2y^2)$
9	27; $9x^4$
10	28, 30; bars 7.2 and 0.6
11	140°; 70°
12	135
13	8.81 cm; 55.4 cm^2
14	46.475 cm^2; 9.49 cm
15	$\frac{2x + 1}{(x + 2)(x - 1)}$
16	$-1.5 \leq x \leq 4$; 6
17	7; $x = 1$ or $-\frac{5}{3}$
18	shown
19	9 m/s; 8 m
20	$n = 10$
21	$CN : NM = 2 : 1$
22	59.9 m; 55.9 m
23	$8n$; e.g. $3^2 - 1^2 = 8$
24	139 cm^2
25	$(x - 3)^2 - 4$; $a = 3$, $b = 4$; (1.5, −4)
26	525 ml

1 Fractions without a calculator

3 marks

1 **IMPROPER FRACTIONS**
$\frac{8}{3} \times \frac{9}{8}$ **M1**

2 **CANCEL AND MULTIPLY**
$= \frac{72}{24}$ **M1**

3 **SIMPLIFY**
$= 3$ **A1**

2 Repeated percentage change

3 marks

4 **MULTIPLIERS**
$1.12 \times 0.88 = 0.9856$ **M1 M1**

ANSWER 1.44% decrease **A1**

3 Prime factors and LCM

3 marks

ANSWER (a) $2^3 \times 3^2 \times 7$ **M1 A1** (b) $2^3 \times 3^2 \times 5 \times 7 = 2520$ **B1**

4 Venn diagram

4 marks

5 **BOTH**
$35 + 28 + 9 - 60 = 12$ **M1**

6 **REGIONS**
French only 23, both 12, Spanish only 16, outside 9 **A2**

ANSWER (b) $\frac{23 + 16}{60} = \frac{39}{60} = \frac{13}{20}$ **B1**

5 Simultaneous equations

3 marks

7 **ELIMINATE y**
$9x + 6y = 48$ and $10x - 6y = 28$; add: $19x = 76$ **M1**

ANSWER $x = 4$ **A1**, $y = 2$ **A1**

6 Bearings

3 marks

8 **(A)**

$12 \sin 65^\circ = 10.87\ldots$ **M1**

ANSWER (a) 10.9 km **A1** (b) $065 + 180 = 245^\circ$ **B1**

7 Average speed

3 marks

9 **TIMES**

$120 \div 80 = 1.5$ h, $90 \div 60 = 1.5$ h **M1**

10 **TOTAL DISTANCE ÷ TOTAL TIME**

$210 \div 3$ **M1**

ANSWER 70 km/h **A1**

8 Expand and factorise

4 marks

ANSWER (a) $12x^2 + 15x - 8x - 10 = 12x^2 + 7x - 10$ **M1 A1** (b) $6xy(3x - 2y^2)$ **B2**

9 Indices

3 marks

ANSWER (a) $\left(\sqrt[4]{81}\right)^3 = 3^3 = 27$ **B1** (b) $27^{2/3} = 9$, $(x^6)^{2/3} = x^4$: $9x^4$ **B2**

10 Histogram both ways

4 marks

11 **(A) FREQUENCY = FD × WIDTH**

$10 < t \le 15$: $5.6 \times 5 = 28$; $20 < t \le 40$: $1.5 \times 20 = 30$ **B1 B1**

12 **(B) FD = FREQUENCY ÷ WIDTH**

$15 < t \le 20$: $36 \div 5 = 7.2$; $40 < t \le 60$: $12 \div 20 = 0.6$ **B1 B1**

11 Tangents and circle theorems

4 marks

13 **(A)**

$OAP = OBP = 90^\circ$ — a tangent meets a radius at 90°; quadrilateral angles add to 360° **M1**

ANSWER (a) $360 - 90 - 90 - 40 = 140^\circ$ **A1** (b) $140 \div 2 = 70^\circ$ — angle at the centre is twice the angle at the circumference **B1 B1**

12 Direct proportion

3 marks

14 **FORMULA**

$y = kx^3$: $40 = 8k \Rightarrow k = 5$ **M1 A1**

ANSWER $5 \times 27 = 135$ **A1**

13 Sine rule and area

5 marks

15 **THIRD ANGLE**

$C = 180 - 38 - 64 = 78^\circ$ **B1**

16 **SINE RULE**

$$\frac{BC}{\sin 38^\circ} = \frac{14}{\sin 78^\circ}$$ **M1**

17 **AREA**

$\frac{1}{2}(14)(8.81\ldots)\sin 64^\circ$ **M1**

ANSWER (a) 8.81 cm **A1** (b) 55.4 cm^2 **A1**

14 Bounds

4 marks

18 **(A)**

8.45×5.5 **M1**

19 **(B) LOWER BOUNDS**

$\sqrt{8.35^2 + 4.5^2}$ **M1**

ANSWER (a) 46.475 cm^2 **A1** (b) 9.49 cm **A1**

15 Algebraic fractions

3 marks

20 **FACTORISE**

$x^2 + x - 2 = (x + 2)(x - 1)$ **M1**

21 **COMMON DENOMINATOR**

$$\frac{3(x - 1) - (x - 4)}{(x + 2)(x - 1)}$$ **M1**

ANSWER $\dfrac{2x + 1}{(x + 2)(x - 1)}$ **A1**

16 Quadratic inequality

4 marks

22 **ROOTS**

$(2x + 3)(x - 4) = 0 \Rightarrow x = -1.5,\ x = 4$ **M1 A1**

ANSWER (a) $-1.5 \le x \le 4$ **A1** (b) $-1, 0, 1, 2, 3, 4$: 6 integers **B1**

17 Functions

4 marks

23 **(A)**

$f(2) = 2,\ g(2) = 7$ **B1**

24 **(B)**

$(3x + 1)^2 - 2 = 14 \Rightarrow (3x + 1)^2 = 16$ **M1**; $3x + 1 = \pm 4$ **M1**

ANSWER $x = 1$ or $x = -\frac{5}{3}$ **A1**

18 Recurring decimal

2 marks

25 **TWO MULTIPLES**

$x = 0.07272\ldots,\ 100x = 7.2727\ldots$ **M1**

26 **SUBTRACT**

$99x = 7.2 \Rightarrow x = \dfrac{72}{990} = \dfrac{4}{55}$ **A1**

19 Kinematics with calculus

5 marks

THE IDEA distance is not displacement: the particle stops and turns back, so add the distance out and the distance back.

27 **(A) VELOCITY**

$v = \dfrac{ds}{dt} = 3t^2 - 12t + 9$; at $t = 4$: $48 - 48 + 9 = 9$ **M1 A1**

28 **(B) AT REST**

$3(t - 1)(t - 3) = 0 \Rightarrow t = 1,\ 3$ **M1**

29 **POSITIONS**

$s(0) = 0,\ s(1) = 4,\ s(3) = 0$: out 4 m, back 4 m **M1**

ANSWER (a) 9 m/s (b) 8 m **A1**

20 Probability — same colour

5 marks

THE IDEA same colour = $WW + BB$; the black counters are $n - 4$. Two answers come out — the condition picks one.

30 **TWO PATHS**

$$\frac{4 \cdot 3 + (n-4)(n-5)}{n(n-1)} = \frac{7}{15}$$ **M1 M1**

31 **SIMPLIFY**

$15(n^2 - 9n + 32) = 7n^2 - 7n \Rightarrow 8n^2 - 128n + 480 = 0$ **M1**

32 **SOLVE**

$n^2 - 16n + 60 = 0 \Rightarrow (n-6)(n-10) = 0$ **M1**

ANSWER $n = 10$ ($n = 6$ gives 2 black, fewer than white) **A1**

21 Vectors in a parallelogram

5 marks

THE IDEA find $\vec{CN}$ and $\vec{CM}$ from the same point C; one is a multiple of the other.

33 **POINTS**

$\vec{OB} = \mathbf{a} + \mathbf{c}$, $\vec{ON} = \frac{2}{3}(\mathbf{a} + \mathbf{c})$, $\vec{OM} = \mathbf{a} + \frac{1}{2}\mathbf{c}$ **M1**

34 $\vec{CN}$

$\frac{2}{3}\mathbf{a} + \frac{2}{3}\mathbf{c} - \mathbf{c} = \frac{2}{3}\mathbf{a} - \frac{1}{3}\mathbf{c}$ **M1**

35 $\vec{CM}$

$\mathbf{a} + \frac{1}{2}\mathbf{c} - \mathbf{c} = \mathbf{a} - \frac{1}{2}\mathbf{c} = \frac{3}{2}\vec{CN}$ **M1**

36 **CONCLUSION**

parallel with a common point C, so C, N, M are collinear **A1**

ANSWER $CN : NM = 2 : 1$ **A1**

22 Two angles of elevation

6 marks

THE IDEA use the non-right-angled triangle ABQ first: its angle at Q is $47 - 32 = 15^\circ$.

37 **ANGLES IN ABQ**
$ABQ = 180 - 47 = 133^\circ$, $AQB = 15^\circ$ **M1**

38 **SINE RULE**
$\dfrac{BQ}{\sin 32^\circ} = \dfrac{40}{\sin 15^\circ} \Rightarrow BQ = 81.90\ldots$ **M1 M1**

39 **RIGHT-ANGLED TRIANGLE BPQ**
$PQ = BQ \sin 47^\circ$ **M1**

ANSWER (a) 59.9 m **A1** (b) $BP = BQ \cos 47^\circ = 55.9$ m **B1**

23 Algebraic proof

4 marks

THE IDEA write two consecutive odd numbers as $2n - 1$ and $2n + 1$.

40 **EXPAND**
$(2n + 1)^2 - (2n - 1)^2 = (4n^2 + 4n + 1) - (4n^2 - 4n + 1)$ **M1 M1**

41 **CONCLUSION**
$= 8n$, which is 8× an integer, so a multiple of 8 **A1**

ANSWER (b) e.g. $3^2 - 1^2 = 8$, not a multiple of 16 **B1**

24 Segment of a circle

5 marks

THE IDEA the perpendicular from O bisects the chord: a 5–12–13 triangle gives the angle.

42 **HALF THE ANGLE**
$\sin\theta = \frac{12}{13} \Rightarrow \theta = 67.38\ldots^\circ$, so $AOB = 134.76\ldots^\circ$ **M1 A1**

43 **SECTOR**
$\frac{134.76\ldots}{360} \times \pi \times 13^2 = 198.74\ldots$ **M1**

44 **TRIANGLE**
$\frac{1}{2} \times 24 \times 5 = 60$ **M1**

ANSWER $198.74\ldots - 60 = 139$ cm^2 **A1**

25 Transformations of a graph

5 marks

THE IDEA the turning point of f is $(3, -4)$; $f(x + a)$ moves it left by a, $+b$ moves it up by b, $f(2x)$ halves the x-coordinate.

45 **(A)**

$(x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$ **M1 A1**

46 **(B)**

turning point $(3 - a,\ -4 + b) = (0, 0)$ **M1**

ANSWER (b) $a = 3$, $b = 4$ **A1** (c) $(1.5, -4)$ **B1**

26 Similar containers

5 marks

THE IDEA area factor → length factor → volume factor; then the 500 ml is the DIFFERENCE between the volumes.

47 **LENGTH FACTOR**

$\sqrt{\frac{100}{64}} = \frac{10}{8} = 1.25$ **M1**

48 **VOLUME FACTOR**

$1.25^3 = 1.953125$ **M1**

49 **DIFFERENCE**

$1.953125V - V = 500$ **M1**

50 **SOLVE**

$0.953125V = 500$ **M1**

ANSWER $V = 525$ ml (3 s.f.) **A1**